AVINYA 2026 — PRAKRITI ECOINNOVATE CHALLENGE | ROUND 1 PROPOSAL

CarbonCampus
Urban Carbon Footprint Calculator

A lightweight, campus-tailored digital tool that turns daily habits into personal emissions insights and automated reduction strategies.

Problem Statement: Urban Carbon Footprint Calculator (Software & Web Development)

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Institution: IIT Guwahati

PROBLEM UNDERSTANDING

The Problem We Are Solving

- **Campuses are emission hotspots: —** universities generate significant GHG emissions through commuting, hostel energy use, canteen food, labs and waste disposal.
- **Quantification is the first step: —** you cannot reduce what you do not measure — but most individuals have no idea what their personal footprint actually is.
- **Existing tools are generic: —** global calculators give broad numbers without campus-specific context (local grid mix, mess menu, shuttle routes, campus waste practices).
- **Awareness does not become action: —** one-time calculators stop at a number; they do not nudge sustained behavior change.

Our Insight

Behavior changes when impact is personal, local, continuous and social. A campus-first tool with localized emission factors, habit tracking and peer engagement converts awareness into measurable, sustained emissions reduction.

MAPPING TO THE PROBLEM STATEMENT

Objectives & Goal Alignment

Objective (from PS)	Build a lightweight, interactive digital tool tailored for university campuses that lets individuals input daily habits, visualize their carbon footprint, and receive automated reduction strategies.
Goal (from PS)	Empower students and faculty with clear, data-driven insights about personal environmental impact, fostering a culture of sustainability and actionable behavioral change across campus.
How CarbonCampus delivers	Activity-based inputs (transport, energy, food, waste) x campus-calibrated emission factors = live footprint dashboard + personalized, automated action plans + hostel/department leaderboards for positive peer pressure.

PROPOSED SOLUTION

CarbonCampus — Solution Overview

CarbonCampus is a Progressive Web App (PWA) — installable, mobile-first, works on low-end devices and poor networks. Three core loops: MEASURE, UNDERSTAND, ACT.

1. MEASURE

Quick daily/weekly habit log: commute mode & distance, electricity & device usage, mess/food choices, single-use plastic and waste. QR-based check-ins for campus shuttles and events. Under 60 seconds per day.

2. UNDERSTAND

Real-time personal dashboard: total kg CO_2e, category-wise breakdown, trend lines, equivalents (trees, km driven), and comparison with campus average and peer cohort.

3. ACT

Automated, personalized reduction strategies ranked by impact and effort (e.g., switch 2 trips/week to shuttle = -18 kg CO_2e/month), weekly nudges, and challenge badges.

Gamification layer: hostel & department leaderboards, green points redeemable for campus perks, monthly campus emission report for administration.

WHAT MAKES IT CLICK

Key Features

Campus-Calibrated Engine

Emission factors tuned to the campus: local grid emission factor (CEA CO2 Baseline Database), mess menu, shuttle fleet, hostel electricity profiles.

Smart Activity Logging

Preset campus templates (hostel day, class day), GPS-assisted trip detection, and one-tap repeat habits — logging in under a minute.

Personalized Action Plans

Rule-based recommendation engine surfaces the top 3 highest-impact swaps per user, updated weekly as habits change.

Visualization That Persuades

CO2e equivalents (trees grown, phone charges, km cycled), category treemaps, trend sparklines, hostel heatmaps.

Challenges & Leaderboards

Hostel vs hostel leagues, 7-day low-carbon streaks, event-specific challenges (e.g., Techniche green week).

Admin Analytics Portal

Aggregated, anonymized campus emission inventory: mobility, energy, food, waste — feeding the institute's sustainability reporting (ISO 14064-1 aligned).

Offline-First PWA

Cached logging and dashboards work offline; syncs when connected. Lightweight (<2 MB app shell).

Privacy By Design

No precise location stored; individual data anonymized in all aggregate views; minimal permissions.

HOW EMISSIONS ARE ESTIMATED

Calculation Methodology

Emissions (kg CO_2e) = Activity Data x Emission Factor (source-calibrated). Four campus-relevant scopes:

Category	User Inputs	Emission Factor Source
Transport	Mode (walk/cycle/shuttle/bike/car/bus/auto), distance, frequency	MoRTH fuel economy data; CEA; DEF rail/bus factors; EV grid factor
Energy	Hostel/lab/device usage hours, appliance type, AC usage	Regional grid emission factor (CEA CO_2 Baseline Database, updated yearly)
Food	Mess meals (veg/non-veg), outside food frequency, food waste level	Published LCA datasets (Poore & Nemecek 2018, Science); Indian meal factors
Waste & Consumption	Single-use plastics, paper, e-waste, packaging	CPCB waste emission factors; EPA WARM model equivalents

Validation approach: pilot with one hostel, cross-check aggregate estimates against institute electricity/transport bills; uncertainty bounds shown to users; factors versioned and auditable in-app.

TECHNICAL FEASIBILITY

Technical Architecture & Stack

Frontend (PWA)

React + TypeScript, Recharts for visualization, IndexedDB for offline logging. Mobile-first, <2 MB shell, installable, works in campus browsers.

Backend API

Node.js (Express/Fastify) REST API: auth (campus SSO / email OTP), habit logs, emission engine, recommendations, leaderboards. Stateless and containerized.

Emission Engine

Versioned emission-factor service: EF = f(activity, campus, year). All factors stored as auditable JSON/DB records with source citations; pluggable per campus.

Database

PostgreSQL: users, logs, factors, teams; Redis: leaderboards & caching. Aggregations pre-computed nightly for dashboards.

Analytics & Notifications

Nightly batch jobs (trend analytics, recommendation refresh, campus reports); email/in-app nudges; admin report generator (PDF/CSV).

Hosting

Dockerized; free-tier/low-cost deployable (Render/Fly.io + managed Postgres). Designed to later move to institute servers for zero hosting cost.

FROM SIGN-UP TO SUSTAINED ACTION

User Journey

1. **Onboard**

   Sign in with campus email; pick hostel, department, year.

2. **Baseline Quiz**

   One-time 2-minute quiz sets the personal baseline footprint.

3. **Daily Log**

   Under 60s habit log; templates auto-fill repeating patterns.

4. **Dashboard**

   See live CO_2e, breakdown, equivalents vs campus average.

5. **Act**

   Get top-3 weekly swaps; accept challenges; earn green points.

6. **Compete**

   Hostel leaderboards, streaks, and campus-wide green weeks.

7. **Report**

   Admin gets monthly anonymized campus emission inventory.

Engagement hooks: streaks, weekly impact digest, redeemable green points (canteen/library perks), and a visible 'campus trees saved' counter powered by the whole community.

SOLUTION INNOVATION — 30% WEIGHTAGE

What Is Innovative Here

Hyper-local calibration

First calculator to parameterize emission factors per campus (grid region, mess menu, shuttle routes) instead of generic global averages.

Action engine, not just a number

Rule-based recommender ranks every possible habit swap by (impact x ease) and delivers exactly 3 per week — behavior-science driven (small wins).

Social infrastructure

Hostel/department leagues convert sustainability into a team sport; peer pressure becomes a decarbonization mechanism.

Institutional feedback loop

Aggregated data doubles as the campus GHG inventory, giving administration a free reporting tool — adoption incentive.

Equivalents that resonate

CO2 is abstract; 'your weekend commute = charging 3,400 phones' is not. Campus-specific equivalents make impact tangible.

Friction-free by design

PWA, offline-first, under-60-second logging, auto-filled templates — engineered for real student attention spans and low-end devices.

TECHNICAL FEASIBILITY — 25% WEIGHTAGE

Why It Is Buildable Now

- **Proven components only: —** web tech, factor-based emission accounting (IPCC Tier-1 style), standard auth — no speculative tech.
- **Open, citable data: —** CEA grid factors, MoRTH fuel economy, CPCB waste factors, published LCA datasets — all public.
- **Reference points: —** builds on lessons from tools like EPA Carbon Footprint Calculator, WWF Footprint, and campus tools (e.g., IITs' own sustainability audits), adapted for Indian campuses.
- **MVP in weeks: —** team can ship logging + engine + dashboard within the hackathon sprint; leaderboards and admin portal follow.
- **Low cost to run: —** free-tier hosting for pilot; institute can self-host later on existing servers.

MVP Scope (Round 2 hackathon)

Week 1: emission engine + factor DB, auth.

Week 2: habit logger + personal dashboard.

Week 3: recommendations + equivalents.

Demo: live hostel pilot with 50+ users and an admin report export.

SCALABILITY & PRACTICALITY — 25% WEIGHTAGE

Scaling Beyond One Campus

Campus -> multi-campus	Emission factors are configuration, not code: onboarding a new college = new factor pack (grid region, mess, transport) + branding.
Student -> city	Same engine scales to wards/cities by swapping campus templates for citizen templates; factor service already region-aware.
Architecture scales	Stateless API + pre-computed aggregates keep costs flat; DB indexes on user/time handle 10k-100k users comfortably.
Sustainable business model	Free for students; institute sustainability-office license; CSR/green-fund sponsorship for challenges; no user-data monetization.
Practical adoption path	Start with Prakriti/eco-club pilots -> hostel leagues -> institute-wide via orientation onboarding -> MoU with other colleges via Techniche network.

SUSTAINABILITY IMPACT — 20% WEIGHTAGE

Sustainability Impact & KPIs

Conservative Impact Model (1 campus, 10,000 users)

If 30% adopt weekly and average just 15% footprint reduction:

Baseline approx 2.5 t CO_2e/person/year -> ~450 t CO_2e avoided annually
= ~20,000 trees' yearly absorption, or 1.8 million EV-km.

Behavioral literature (feedback + social comparison) supports 5-20% reductions — we target the low end.

KPIs We Will Track

MAU & logging streak rate
% users acting on >=1 recommendation
Average footprint change vs baseline (monthly)
Hostel participation in leagues
Admin report usage for sustainability filings

SDG Alignment

SDG 4 (Quality Education - sustainability literacy) | SDG 7 (Affordable & Clean Energy - informed energy use) | SDG 11 (Sustainable Cities & Communities) | SDG 12 (Responsible Consumption & Production) | SDG 13 (Climate Action - personal & institutional decarbonization)

EXECUTION PLAN

Roadmap & Team

Phase 1 - MVP (Aug-Sep 2026)	Emission engine + factor database, PWA logging & dashboard, top-3 recommendations, pilot with 1-2 hostels during Techniche.
Phase 2 - Campus Rollout (Oct-Dec 2026)	Leaderboards & challenges, admin analytics portal, campus emission report v1, integration with eco-club events.
Phase 3 - Scale (2027)	Multi-campus factor packs, API for other colleges, open-source emission-factor registry, institutional licensing.

Team

[Mandavi- Full-stack dev] • [Gaurav - Data/emission modelling] • [Gaurav & Mandavi - UX & content]

Measure it. Shrink it. Repeat.

CarbonCampus turns every student into a climate actor - one habit at a time.

Thank you. We look forward to presenting at IIT Guwahati during Techniche.

Avinya 2026 - Prakriti EcoInnovate Challenge | EcoGenesis

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Demo link: